NutriCare AI

Smart Nutrition Assistant

How to Run the Web Application

Step-by-step guide - October 05, 2026

The app runs TWO local services and then opens in your browser:

```
  Service 1  Node.js backend      port 8000  (serves the website + API)
  Service 2  Python RAG / ML      port 5000  (AI chat, food detection,
                                             recommendations, 15-day chart)

  Browser    http://127.0.0.1:8000
```

1. Prerequisites

Requirement	Used for	Check in a terminal
Node.js 18+ (project runs on v24)	Web server on port 8000	node -v
Python 3.13	RAG / ML service on port 5000	python --version
Ollama with llama3.2	AI answers in the chat bot	ollama list
Python packages	RAG service dependencies	pip install -r backend\requirements.txt

NOTE

Windows note: if 'node' or 'python' is not recognised, install them first from nodejs.org and python.org, then reopen the terminal.

2. First-time setup (run once)

Open a terminal in the project folder (D:\JNEC\Nutricare AI) and run:

```
cd /d "D:\JNEC\Nutricare AI"
npm install
pip install -r backend\requirements.txt
ollama pull llama3.2     ; only if 'ollama list' does not show it
```

3. Step 1 - Start the two services

Option A (easiest): use the start script

Double-click start.bat in the project folder. It opens two windows:

- NutriCare - Node (8000) - runs npm start
- NutriCare - Python RAG (5000) - runs python rag_service.py

Wait for the Python window to show 'Started server process' and finish loading the knowledge base (about 10-30 seconds on first start).

Option B: start manually in two terminals

```
:: Terminal 1 - Node backend (port 8000)
```

```
cd /d "D:\JNEC\Nutricare AI\backend"
npm start

:: Terminal 2 - Python RAG service (port 5000)
cd /d "D:\JNEC\Nutricare AI\backend"
python rag_service.py
```

WARNING - the most common mistake

Port 5000 must run backend\rag_service.py. Do NOT start services\ml_api.py or app.py on port 5000 - they do not have the /rag-chat, /food-recommend or /plan/15-day routes. If the wrong file is running, the chat and the food chart fail with the message: 'Food chart service returned an error.' Fix: close that window and run 'python rag_service.py' instead (or just use start.bat).

4. Step 2 - Open the application

Once both windows are running, open this URL in your browser:

```
http://127.0.0.1:8000
```

You land on the NutriCare AI dashboard. The navbar shows your name, your BMI (top bar) and the hamburger menu on phones.

5. Step 3 - Verify everything works

URL	What you should see
http://127.0.0.1:8000/api/health	JSON with status ok (Node is up)
http://127.0.0.1:8000/api/ml/health	python_ml_service: online, doc_count 151210
http://127.0.0.1:8000/Main/index.html	Dashboard loads (HTTP 200)
http://127.0.0.1:5000/	NutriCare AI RAG Service, status online

Quick check from the command line:

```
curl http://127.0.0.1:8000/api/health
curl http://127.0.0.1:5000/
```

6. Step 4 - Log in

TEST ACCOUNT

Email: krishna@test.com Password: 123456

No OTP is required - registration uses the password form only (Login -> Register).

7. Step 5 - Use the features

- Food image - Upload & Analyze -> Food Image: 43-class AI detection with calories, protein, sugar, sodium, risk badges.
- Reports - Upload a blood sugar or health report for AI analysis.
- Chat bot - Ask anything in the NutriCare AI Assistant. Answers use the 151k-document knowledge base and your profile (about 5-60 s per answer - Ollama generation time is normal).
- Foods I Can Eat - My Food Plan -> Foods I Can Eat: foods sorted into eat freely / moderate / limit for your BMI and conditions.

- 15-day food chart - My Food Plan -> Create 15-Day Food Chart: 15 days x 8 meals sized to your calorie target, with Download CSV and Print / PDF buttons.
- Step tracker - Live Step Tracker uses your phone's motion sensor.
- Log out - Click the avatar in the top bar.
- Admin - Open http://127.0.0.1:8000/Admin/admin.html for user lists and database info.

On a phone the sidebar collapses behind the Hamburger (three lines) button, and the 15-day chart scrolls sideways inside its own box.

8. Key API endpoints

Method	Endpoint	Purpose
POST	/api/chat	RAG chat with user profile
POST	/api/recommend	Foods you can eat (3 groups)
POST	/api/plan	15-day food chart
POST	/api/analyze-food	Food image to nutrition
GET	/api/health	Node health
GET	/api/ml/health	Python + Chroma health
POST	:5000/rag-chat	RAG chat (internal)
POST	:5000/food-recommend	Recommendations (internal)
POST	:5000/plan/15-day	15-day chart (internal)

9. Troubleshooting

Symptom	Cause	Fix
'Food chart service returned an error.'	Wrong Python file on port 5000	Stop it and run python rag_service.py (use start.bat)
Port already in use / EADDRINUSE	An old instance is still running	Close the old terminal windows, then start again
Chat answers slowly (5-60 s)	Ollama generation time (normal)	Wait for the answer; keep llama3.2 installed
Chat says no context / cannot answer	Ollama is not running	Start 'ollama serve' and check 'ollama list'
Python window closes immediately	Missing packages	Run 'pip install -r backend\requirements.txt'
First chart click is slow	Models + knowledge base loading (cold start)	Only the first call; later calls take under 1 s
Page loads but files 404	Wrong address opened	Always start at http://127.0.0.1:8000

10. Stop and restart

- Stop - Close both terminal windows (or press Ctrl+C in each).
- Restart - Run start.bat again.

DISCLAIMER

NutriCare AI is intended for research and educational purposes. Food recommendations come from a dataset classifier and are NOT a substitute for professional medical diagnosis or treatment.